

RNA co-Immunoprecipitation

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Day 0: Plate Cells (for cos-1 cells, split cell ~8hr before transfection)

- COS-1 cells: 6e5 cells/poly lysine-coated 60m plate; 293T: 2e6 cells/60mm plate.
- Use 4% lysate for TCL IP ($=4.72 \times 10^4$ cell equivalent, 25 μ L out of 620.5 μ L lysate)

Day 0: Transfect Cells

8 hours after plating cells, transfect each plate with desired plasmid(s).

Before starting:

- Label 1 1.5 mL Eppies per sample.
 - Thaw DNA at 55°C in Thermomixer and rotate at >1000 RPM for 15 minutes to resuspend.
 - Thaw PEI (MANUFACTURER, CATALOGUE NUMBER) at 55°C in Thermomixer and rotate at >1000 RPM for 15 minutes to resuspend.
1. For each transfection condition, dilute DNA in 500 μ L pre-warmed, serum-free medium.
 2. Let sit 5 minutes at room temperature (RT).
 3. Add 20 μ L of pre-warmed PEI (more than 15min at 55°C) to DNA/ serum-media, incubate at RT 15 minutes.
 4. Change media on cells to 3 mL of transfection media.
 5. Add transfection mix dropwise to cells while gently rocking the plate.

Day 1: Change media (when you culture cells more than 24 hrs after transfection)

Before starting:

- Warm PBS + MgCl₂/CaCl₂
- Warm culture media (for 293Ts and Cos-1, 500 mLs DMEM + 10% FBS + 5 mLs penicillin/streptomycin)

~14 hours post transfection, rinse cells twice with 3 mL warmed PBS containing CaCl₂ and MgCl₂ to remove untransfected DNA, then add 3.5 mL fresh media.

Day 2: HARVEST (1xNMPH harvest), Gradient fractionation, IPs

Harvest lysate in NMPH (dissociating ribosomes and releasing the nascent chain):

Before starting:

- Label and pre-chill 3 1.5 mL Eppie tubes per sample.
- Pre-chill benchtop Allegra centrifuge to 4°C.
- Pre-chill a microcentrifuge to 4°C.
- Pre-chill 1 mL syringes with 20 gauge needles attached.
- Prepare a "cell scraper" – an aspirator pipette with a rubber policeman scraper at the end.
- Thaw an aliquot of Puromycin (need 26.5 μ L for each sample) – Puromycin should not undergo multiple freeze-thaw cycles.
- Make sure you have 144 μ L of 2M KCl NN+M (high salt) per sample, keep chilled.

1. Rinse cells with 3mL ice-cold PBS containing CaCl₂ and MgCl₂ twice to remove untransfected DNA.

2. Add 500 μL of 1x Lysis Buffer NN+M (+ 1:100 Protease Inhibitor and 1:1000 RNaseOUT) per plate, scrape lysate, transfer to chilled tube.
3. Shear lysed cells 20-times through a chilled 20-gauge needle.
4. Spin lysed cells at 1000 RPM for 10 minutes at 4°C in tabletop Allegra, then move supernatant to a new tube.
5. Spin supernatant at 14k RPM 1 minute 4°C in microcentrifuge.
6. Transfer 450 μL of lysate to a new tube.
7. Warm up lysate and Puromycin for 5 minutes at 26°C.
8. Add 26.5 μL Puromycin to 450 μL of lysate at final concentration of 1mM (stock concentration of Puromycin is 18mM)
9. Incubate for 10 minutes at 26°C.
10. Add 144 μL 2M high salt NN+M at final concentration 0.5M KCl (620.5 μL lysate final volume).
11. Incubate for 10 min at 26°C
12. Place tubes on ice for at least 10 minutes.
13. Centrifuge at max speed for 1 min in a 4°C microcentrifuge.

Sucrose Gradient: 5mL Gradient

Gradient A) 10-80% Gradient: suitable to see the assembly intermediate pattern in wide range

Use (MANUFACTURER, CATALOGUE NUMBER, DIMENSIONS) ultracentrifuge tubes

10/15/40/50/60/70/80% (w/v) sucrose solution containing 1x buffer (80% at the bottom)

675 μL each layer x 7 layers

Load 200 μL of lysate on top of the gradient

Centrifuge at 45K rpm for 45 min at 4°C

Fractionation 175 μL x ~27 fractions, plus the pellet (fraction 28).

Fractions 1-18 are used for individual IPs, fractions 19-21, 22-24, 25-27 are combined and then used for IP.

Fraction 28 is used individually.

Gradient B) 5-30% Gradient: suitable to separate 10S and 80S

Use (MANUFACTURER, CATALOGUE NUMBER, DIMENSIONS) ultracentrifuge tubes

5/10/15/20/25/30% (w/v) sucrose solution containing 1x buffer

Use

790 μL each layer x 6 layers

Load 200 μL of lysate on top of the gradient

Centrifuge at 45K rpm for 90 min at 4°C

Fractionation 240 μL x ~20 fractions or 480 μL x ~10 fractions

All fractions are used for separately for IP

Gradient IPs

Before starting:

- Make up 1x NN IP wash buffer plus detergent (see end of protocol).
- Make up IP beads (do this at the RNA bench with the RNase-free magnet):
 - Aliquot the amount of protein G Dynal beads (MANUFACTURER, CATALOGUE NUMBER) needed (20 μL bead slurry per IP)
 - Wash beads three times with 1mL 1x NN IP wash buffer plus detergent.
 - Label tubes for each IP and place on ice.
- Make up antibody plus IP buffer master mixes (see Methods for antibodies and amounts used).
- Obtain filter tips for ALL pipetting.

Gradient A) 10-80% Gradient

21 IPs=IP from each fraction 1-18 + IPs from combined fractions 19-21, 22-24, 25-27

1. Add 100 μ L IP buffer/ antibody to the new 1.5 mL Eppie tube on ice.
2. Add lysate:
 - a. Add 120 μ L lysate to the tube for each fraction 1-18.
 - b. Add 180 μ L lysate to the tube for combined fraction 19-21, 22-24, 25-27.
3. Add 260 μ L of 1xNN buffer plus detergent (total 360 μ L buffer to 120 μ L of lysate, 4 fold dilution).
4. Incubate on ice for 30 minutes.
5. Add 20 μ L IP beads.
6. Rotate on an RNase-free rotor for 1 hr at 4°C.

Gradient B) 5-30% Gradient

10 IPs=IP from each fraction 1-10 or neighboring combined fractions 1+2, 3+4, 5+6,...19+20

1. Add 100 μ L IP buffer/antibody to the new 1.5 mL Eppie tube on ice.
2. Add 120 μ L lysate to the tube.
3. Add 260 μ L of 1xNN buffer plus detergent (total 360 μ L buffer to 120 μ L of lysate, 4 fold dilution).
4. Incubate on ice for 30 minutes.
5. Add 20 μ L IP beads.
6. Rotate on an RNase-free rotor for 1 hr at 4°C.

Total Cell Lysate (TCL) IPs:

1. Add 100 μ L buffer and antibody to the new 1.5ml Eppie tube on ice.
2. Add 25 μ L TCL to the tube.
3. Add 260 μ L of 1xNN buffer plus detergent (total 360 μ L buffer to 25 μ L of lysate, 15.4 fold dilution).
4. Incubate on ice for 30 minutes.
5. Add 20 μ L IP beads.
6. Rotate on an RNase-free rotor for 1hr at 4°C.

IP washes:

Before starting:

- Obtain 4 mLs 1xNN buffer plus detergent for each sample, aliquot necessary volume into a Falcon tube and keep chilled (see end of protocol).
 - Obtain 1 mL 1xNN buffer no detergent for each sample, aliquot necessary volume into a Falcon tube and keep chilled (see end of protocol).
 - Obtain RNase-free magnets (MANUFACTURER, CATALOGUE NUMBER).
 - Fill two RNase-free tube racks (MANUFACTURER, CATALOGUE NUMBER) with ice, one with the tube-holder lid, one without – KEEP TUBES ON ICE BETWEEN WASHES.
 - NOTE: Samples should be processed in the following order: Non-Immunes (NI), then Immunes (I), then TCL IPs. This will minimize contamination by going from low levels of RNA to higher levels.
 - Fill a designated RNA ice bucket with ice, changing gloves afterwards. This ice can be used to replace the melting ice in the grey trays. NOTE: Change ice frequently (i.e., between 16-set washes) to prevent water from contaminating samples.
 - Throughout protocol, open only 4 tubes at any time.
 - Wipe down the RNA bench and RNA bench Pipetmen with RNase-ZAP (MANUFACTURER, CATALOGUE NUMBER) or similar reagent.
 - Obtain filter tips for ALL pipetting.
1. Remove tubes from the rotator and place on ice in a grey tube rack with tube-holder lid.
 2. For 16 samples at a time, place tubes in the magnet and allow beads to pellet completely.
 3. Invert to collect remaining beads from the cap and allow to pellet again.
 4. **If moving from NI to I or I to TCL IPs, change aspirator tip here.**
 5. Aspirate off supernatant from the tube and cap.

6. Proceeding in the same sample order, go back and aspirate any liquid that has collected after the first aspiration.
7. Remove the magnet from the tube holder and place holder with tubes on the lidless grey ice tray.

Detergent washes:

8. Using an RNase-free repeater, add 1 mL cold IP wash buffer to each tube (opening four tubes at a time, then recapping).
9. Invert tubes in holder to thoroughly resuspend beads, then rotate at RT for 1 minute.
10. Place tubes back in the magnet and allow beads to pellet completely.
11. Invert once to collect remaining beads from the cap and allow to pellet again.
12. Aspirate off supernatant from the tube and cap.
13. Proceeding in the same sample order, go back and aspirate any liquid that has collected after the first aspiration.
14. Repeat three more times for a total of four washes.

Non-detergent wash (proteinase K compatible = no EDTA):

15. Using a filter tip on a P1000 pipetman, add 1 mL of cold non-detergent wash buffer to each tube (again opening four tubes at a time, then recapping).
16. Invert tubes in holder to thoroughly resuspend beads, then rotate at RT for 1 minute.
17. Place tubes in the magnet and allow beads to pellet completely.
18. Invert once to collect remaining beads from the cap and allow to pellet again.
19. Aspirate off supernatant from the tube and cap.
20. Proceeding in the same sample order, go back and aspirate any liquid that has collected after the first aspiration.
21. Place samples back in the ice tray with the tube-holder lid and proceed with washing the next set of 16 samples.

Proteinase K reaction and TRIZOL:

Before starting:

- Make up 1.5 mL Eppie tubes for TCL and VLP from each sample.
- Make up non-detergent 1x NN buffer (see end of protocol).
- Obtain 1 mL Trizol (Ambion, cat# 15590026) for each sample (make an aliquot of the needed volume + 10%, being careful not to draw from the bottom of the bottle).
- Preheat RNA bench heat-block or water bath to 55°C.
- Obtain 20 mg/mL Proteinase K (Sigma P4850).

1. Make up Proteinase K elution buffer

To make 1 ml (for approx. 10 samples):

975 µL of non-detergent 1x NN (see end of protocol)
 15 µL Proteinase K
 10 µL 10% SDS

2. Add elution buffer to samples:

- a. Gradient IPs: remove last wash and add 100 µL proteinase K elution buffer to beads.
- b. Gradient IP input: Take 12 µL from gradient fractions 1-18 and 28 or 18 µL from combined gradient fractions 19-21, 22-24, 25-27 (10% gradient IP input)
- c. TCL input: take 2.5 µL of TCL (10% TCL IP input)

3. Add proteinase K elution buffer to equal 100 µL.
4. Incubate at 55°C for 30 minutes.
5. Add 1 ml TRIZOL. Invert to mix well and incubate at RT for 5 minutes.

- Store at -80°C if stopping.

If you want to stop the experiment, stop at this point. RNA is not stable in isopropanol.
Next stop point is the end of cDNA reaction.

Day 3: RNA isolation and Reverse Transcription reaction

RNA isolation with TRIZOL:

Before starting RNA isolation and DNase treatment:

- Obtain 5 µL Glycoblue (Invitrogen AM9515 or AM9516) per sample.
- Obtain 100 µL BCP (BP 151 phase separation reagent, Molecular Research Center, CATALOGUE NUMBER) per sample.
- Prepare 0.7 mL of ice cold isopropanol per sample (store at -20°C until ready for use).
- Prepare 1 mL of ice cold 75% ethanol per sample with PCR grade or DEPC-treated water (store at -20°C until ready for use).
- Prepare 51 µL of 1xDNase buffer (Invitrogen cat# 18068-015, prepare with DEPC water) containing 1:1000 RNaseOut and 2U (1 µL) DNase (Invitrogen cat# 18068-015) per sample.
- Heat a clean-room heat block or water bath to 25°C.
- Heat a clean-room heat block or water bath to 65°C.
- Obtain 1.5 µL of 25mM EDTA per sample.
- Pre-chill an RNA-designated centrifuge.
- Clean benchtop, Pipetmen, and all equipment with RNase-ZAP (MANUFACTURER, CATALOGUE NUMBER) or similar reagent.

1. If samples are frozen, thaw at room temperature and set on ice. Invert many times to promote lysis of material.
2. Spin down briefly to collect inverted liquid.
3. Add 75 µg of glycoblue (5 µL) and mix well by inverting tube.
4. Spin down briefly to collect inverted liquid.
5. Add 100 µL of BCP.
6. Vortex for 10-15 seconds until emulsion forms, repeat 2 times.
7. Let sit on ice for 2-3 minutes.
8. Spin 12,000xg 15 minutes at 4°C in chilled RNA-designated microcentrifuge.
9. While tubes are spinning, for each sample transfer 0.7 mL of ice cold isopropanol to a fresh labeled tube. Set tubes on ice.
10. With a 1 mL Pipetman, transfer ~0.6 mL of the aqueous phase from the spun sample directly to the tube with the isopropanol. Keep all samples on ice.
11. Mix well by inverting at least 5 times.
12. Transfer to -20°C for at least 1 hour.
13. Spin at full-speed (14K RPM) for 30 minutes in cold room or in chilled benchtop centrifuge, then carefully use a 1 mL pipette to remove the supernatant without touching the pellet (tip the tube).
14. Wash pellet with 1 mL of 75% ice cold EtOH by inverting several times.
15. Spin full speed (14K RPM) for 10 minutes in chilled benchtop centrifuge.
16. Aspirate most of the EtOH.
17. Spin full speed (14K RPM) for 1 minute in chilled benchtop centrifuge, then pipet to remove the remaining EtOH (with P200).

18. MOVE samples to clean room: everything will be in a clean room or clean area after this

19. Allow RNA pellets to dry with open lids for a few minutes (~3). As pellets dry, close tube to prevent over-drying.

DNase I treatment (@ clean room):

1. Add **51 µL of 1xDNase buffer** (made with DEPC water) **containing 1:1000 RNaseOut and 2U DNase (1 µL)**, by pipetting bead of water on sidewall and letting it slide down to pellet. Be sure pellet ends up in water, sometimes they can end up on sidewall (static effect).
2. Tap samples to resuspend pellet. Spin down briefly.
3. Finish mixing with pipet 5 times, running the bead of water over tube sidewall 5 times while doing this to ensure the entire RNA pellet resuspends.
4. Incubate tubes at 25°C for 15 minutes.
5. Add 1.5 µL of 25mM EDTA, mix well, spin.
6. Heat sample at 65°C for 10 mins to inactivate DNase I (follow the manufacture protocol).

RT reaction to make cDNA from RNA (in a clean room or clean area):

Before starting:

- Pre-heat a water bath to 42°C.
- Pre-heat a heat block to 85°C.
- Label 1.5 mL Eppies for one +RT and one –RT control reaction per sample, plus a single no-RNA control.

1. Make cDNA master mix (iScript Advanced cDNA synthesis kit, 172-5037, BIO-RAD), including one no-RNA control (total).

+RT samples and – RNA control rxn (one total)

	x1
PCR grade H ₂ O	10 µL
5X Script reaction mix	4 µL
20X RT enzyme mix	1 µL
(Sample: RNA	5 µL)
total:	15 (20) µL

2. For each sample, aliquot 15 µL master mix into Eppie tubes.
3. Add 5 µL of RNA to each Eppie tube (total 20 µL per reaction for each duplicate).
4. For RNA negative control: aliquot 15 µL master mix into Eppie tube.
5. Add 5 µL DEPC H₂O to this tube: (for no-RNA control..... ensures purity of cDNA reagents)
6. Make cDNA master mix for RT- controls (iScript Advanced cDNA synthesis kit, 172-5037, BIO-RAD) – this does not need to be done on every sample, but should be done for TCLs and a few of the gradient IP samples.

RT- controls to test the purity of the RNA

	x1
PCR grad H ₂ O	15 µL
(Sample: RNA	5 µL)
total:	15 (20) µL

7. Aliquot 15 μL into Eppie tubes.
8. Add 5 μL RNA to each tube (these are the RT- controls, which measure the DNA contamination in the RNA samples).

RT reaction

9. Incubate samples at 42°C for 30 minutes in the preheated water bath.
10. Incubate samples at 85°C for 5 minutes in the preheated heat block.
11. **Dilute cDNA 1:3** (transfer to new tubes – add 51 μL DEPC H₂O + 17 μL cDNA) – these can be stored at -80°C ; but better to immediately proceed to qPCR if you can.

qPCR oligos:

- Oligo information is available in methods.
- When oligos arrive, dilute to 200 μM .
- Mix 200 μM forward and reverse oligos at 1:1 to make 100 μM stock.
- Before use, make 10 μM oligo mix.

qPCR standard for VIB or 1522 genome:

NOTE: always work with standards OUTSIDE of the clean room or clean area!!!

- Initial standard dilution is $1\text{e}8$ copies/ $2\mu\text{L}$ – make many 20 μL of aliquots of this dilution, stored at -80°C .
- Thaw each aliquot only once, use freshly thawed aliquot for each dilution series
- Prepare standards at 1:10 dilutions ranging from $1\text{e}1$ - $1\text{e}8$ copies/ $2\mu\text{L}$ (2 μL =per well), duplicated – add 5 μL each dilution to 45 μL DEPC water to make the next dilution
- Dilution series are only one time use, do not re-freeze for another use.

qPCR plate setup:

All the qPCR reagents are stored in the clean room, and this is where aliquoting is done. Standards are set up outside the clean room.

Before starting:

- Make up 2 25 μL qPCR reactions per sample – set up in clean room.
- Prepare your standard curve (see above) – set up outside the clean room.
- Prepare DEPC water (in the clean room boxes, aliquots made in the hood).
- Thaw Sybr green (MANUFACTURER, cat# 170-8884) and keep on ice, protected from the light.
- Obtain or make up 10 μM oligos.
- NOTE: duplicate qPCR reactions are set up in Eppies first, then aliquoted to plate (include 16 wells of master mix below for duplicate standard curve to after sample qPCRs have been set up).
- Obtain a 96-well plate.

qPCR mix:

	x1
DEPC	9.75 μL
2xSYBR	12.5 μL
10 μM oligos mix	0.75 μL
(Sample: 4 fold diluted cDNA	2 μL)
total:	23 (25) μL

1. Make up duplicate reactions, then aliquot into a clean 96-well plate, being careful not to introduce bubbles. Wells for standards should be in the top two rows, stacked. These will be completed outside the clean room.
2. Add 2 μ Ls of diluted cDNA to each non-standard sample well.
3. Place paper towel over plate in clean room, move to another bench to add prepared standards.
4. Load 2 μ L of each standard per well in duplicate.
5. Seal cover carefully with a foil from the RNA bench drawer, using a sealer from the same drawer to ensure a tight seal. Pay careful attention to the edges.
6. Centrifuge plate briefly to 500 RPM in a tabletop Allegra.
7. Plate is now ready to be set in qPCR machine, with well A1 on the top left and the notch at the bottom left.

qPCR program setting (for gRNA analysis)

1. 95°C, 30 seconds
2. 95°C, 15 seconds
3. 60°C, 15 seconds
4. 72°C, 30 seconds +10 sec for data collection
5. Back to 2, 35 cycles
6. 55°C, 1 min
7. 55°C- T_m measuring

BUFFER RECIPES (below):

Lysis buffers:

Tris is helpful if you need to add calcium or manganese, b/c these will precipitate phosphate (PBS); otherwise, phosphate is preferable. But need to have a good buffering system (Tris pH 7.4) for coIPs.

10X NP40 Normal salts + MgCl₂ (NN+M)

0.1M Tris Hcl (pH 7.4)
 1.0M NaCl
 0.5M KCl
 10mM MgCl₂
 6.25% NP40

1X NP40 Normal salt + MgCl₂ (NN+M high salt 2M KCl)

10mM Tris HCl (pH 7.4)
 0.1M NaCl
 2M KCl
 1mM MgCl₂
 0.625% NP40

IP Wash buffers: USE 1xNN buffer set for NMPH harvested lysate

***Use 1x NN non-det. buffer for WB IP wash**

***Use 1x NN non-det. IP wash (Prot.K –compatible) for RNA analysis wash**

1x NN IP wash buffer 1x NN Non-det. IP wash buffer 1x NN non-det. IP wash (Prot.K –compatible)

10mM Tris-HCl, pH 7.4
0.1M NaCl
50mM KAc
0.625% NP40

10mM Tris-HCl, pH 7.4
0.1M NaCl
50mM KAc
100mM Tris Ac , pH 8.0

10mM Tris-Ac, pH 8.0
0.1M NaCl